Good Data, Bad Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected data-quality audit.

Check the schema, log a correction, inspect duplicates, audit the new flight log and then inspect follow-up evidence.

Keep one habit: Keep values, units, record identity and correction reasons together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confidence Is Not **Correctness**. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Recommendations

Confidence Is Not Correctness

Save New Examples for Testing

W1 · Apply the RampLog schema

> Use the supplied reference and distinguish missing travel from measured zero.

Your evidence record

- RampLog reference: unique trial ID, height as whole cm 1–10, travel as whole cm 0–10, and original source card. Raw records: T1=(4,3,C1), T2=(60,5,C2), T3=(5,blank,C3), T4=(6,0,C4), plus an identical exported T1 row. C1 confirms one T1; C2 confirms height 6 cm and travel 5 cm; C3 has no travel observation; C4 confirms measured zero. The task counts travel at least 4 cm.

- Use the RampLog **schema**. State the issue or valid status for height 60, travel blank and travel 0. Give the range allowed for each numeric field and explain why a blank and zero require different treatment.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Write a traceable correction

> Name old and new values, source, reason, reviewer and date.

Your evidence record

- C2 says T2 height is 6 cm and travel is 5 cm; the export says height 60 cm. Write a change-log entry with ID, field, old and new values with units, **evidence**, reason, reviewer and date. Use a fictional reviewer and date.

  Explain where the raw export is kept.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Distinguish duplicate rows from separate trials

> Use IDs and source identity rather than equal measurements alone.

Your evidence record

- The export repeats T1 with the same source C1, which confirms one trial. A different trial T5 also has height 4 and travel 3, with its own source C5 confirming a separate trial.

  Explain which rows count once and why T5 must remain separate.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Audit an independent paper-plane log

> Keep one first attempt before hints or feedback.

Read and explain independently

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Audit an independent paper-plane log

MAIN LESSON · SECTION 16

W5 · Inspect follow-up evidence after the first flight-log attempt

> Retain the original W4 report while investigating a new conflict or a separate flight's conversion.

Your evidence record

- Two signed cards disagree about the same F3 flight: one says 9 cm and the other 11 cm. Neither is established as the **correction**. What can you report for the at-least-10-cm question?

  Separately, a source for a new F5 flight explicitly records 100 mm. Convert it to centimetres and state the conversion **evidence** to retain.

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Audit the sorter's example cards and add a reason for including each field and condition.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Write a traceable **correction** for Data quality and documented repair. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

What needs checking? May 130 silently become 13?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 21 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

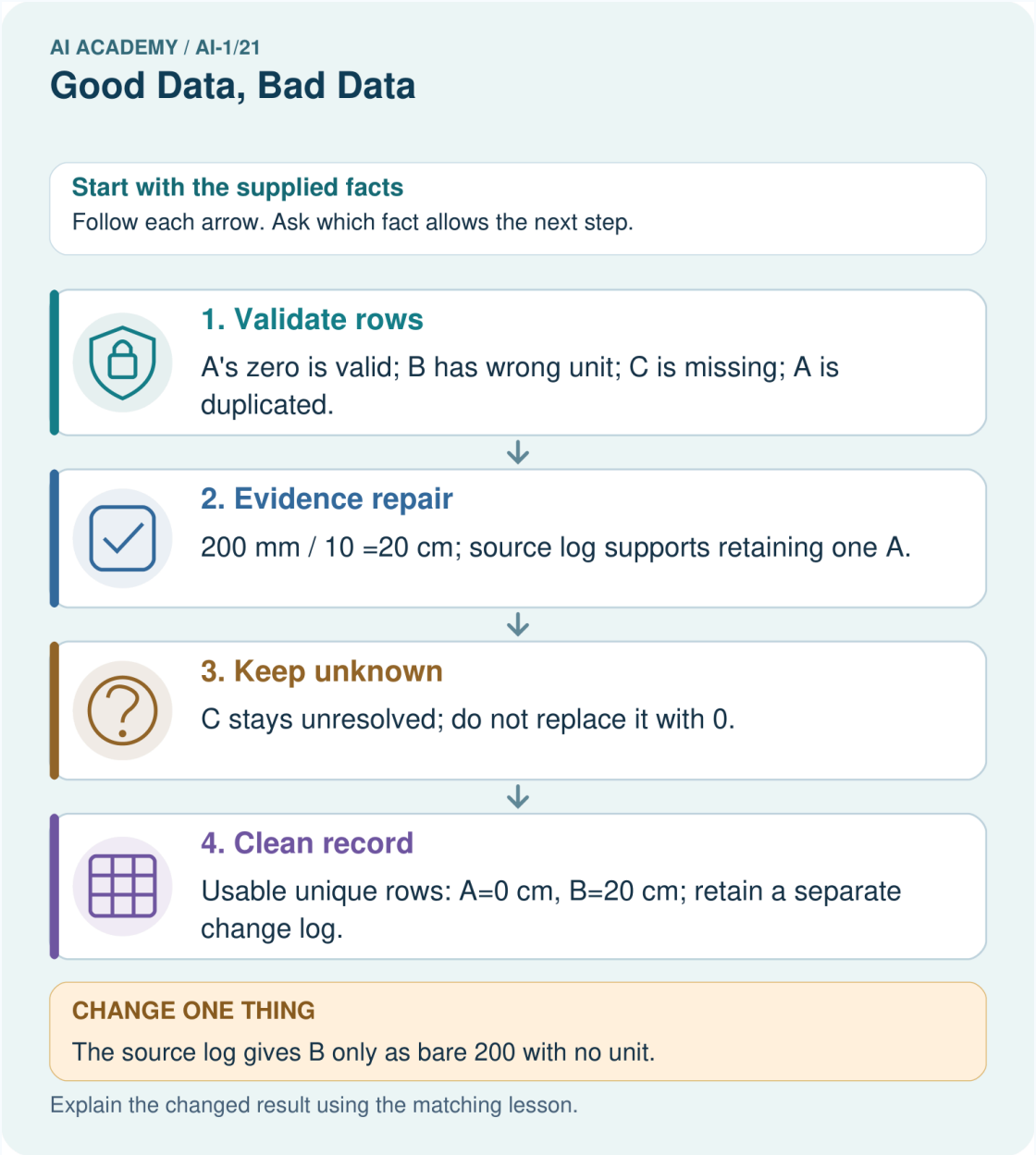
> Keep the explanation and the support you used.

Use this when helpful

A fictional toy-car distance table needs cleaning before use.

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

> **My picture-based explanation and changed-case prediction**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy-car distance table needs cleaning before use.
- Which labelled step matches this detail? Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.
- 1. Clean record

- 2. **Evidence** repair
- 3. Keep unknown
- 4. Validate rows
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> |---|
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▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Clean record Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Cleaning is not permission to make inconvenient values plausible.

The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

Changed-case question

- Changed-case practice: The source log gives B only as bare 200 with no unit. Which conclusion is supported by the supplied facts?
- 1. B remains unresolved rather than becoming 20 cm.
- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: B remains unresolved rather than becoming 20 cm. Without a confirmed unit, division by 10 would be an unsupported assumption.

Repair and save your work

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Review Confidence Is Not **Correctness** (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
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> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

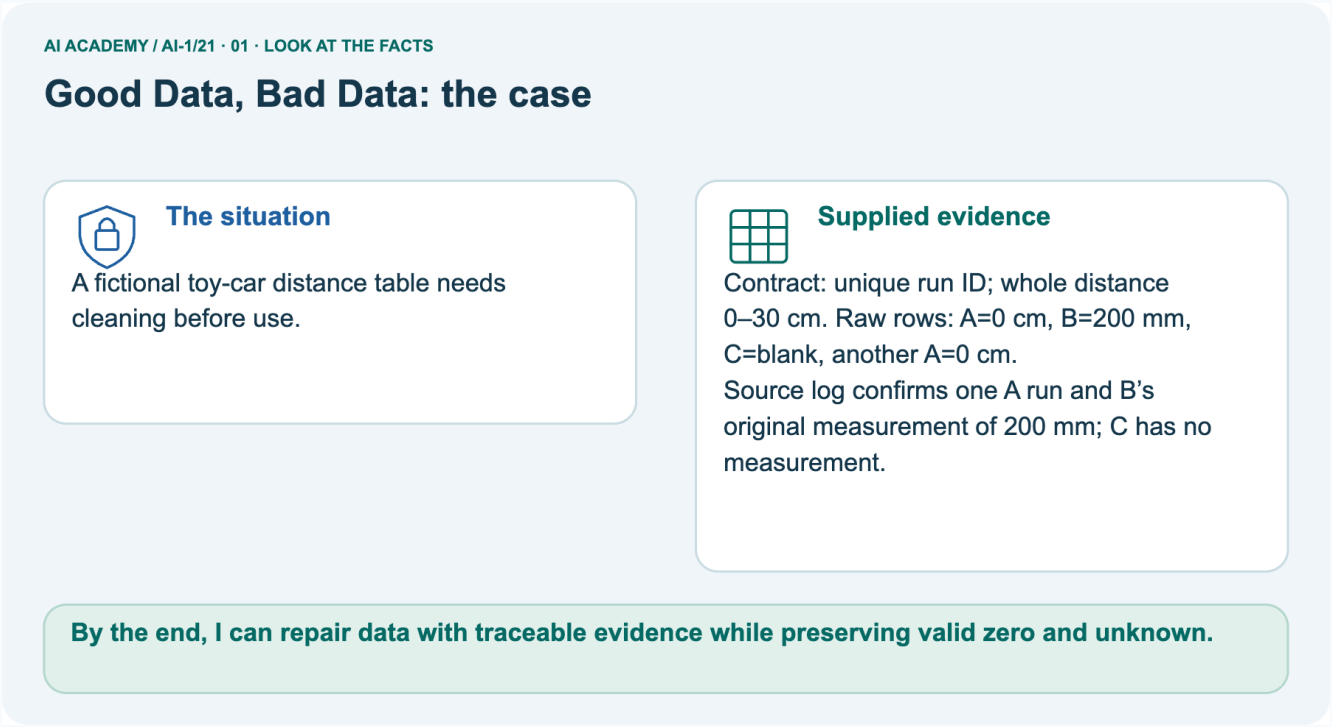
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Good Data, Bad Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

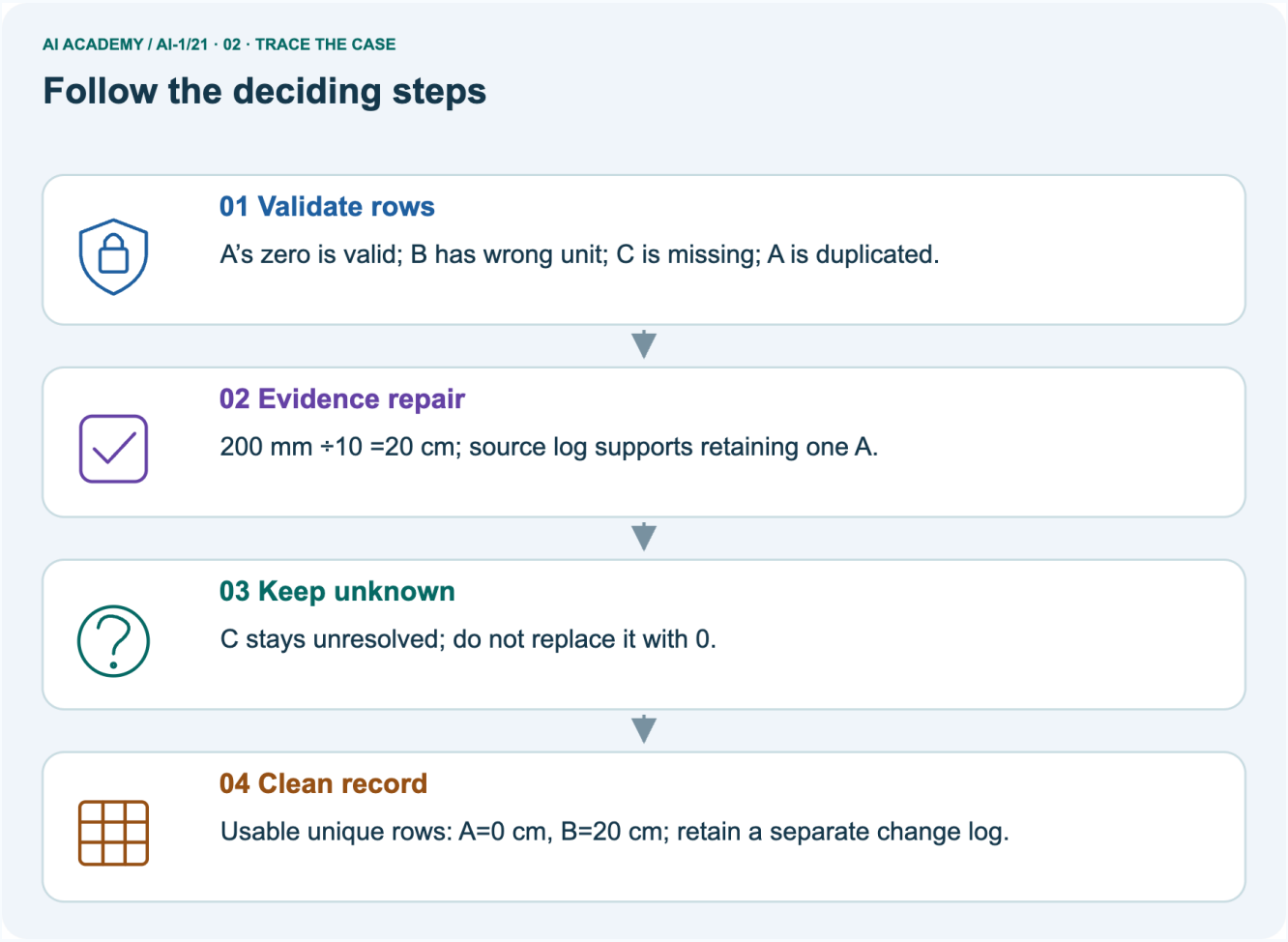
A fictional toy-car distance table needs cleaning before use. Supplied facts: Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate rows: A's zero is valid; B has wrong unit; C is missing; A is **duplicate**d. → **Evidence** repair: 200 mm ÷10 =20 cm; source log supports retaining one A. → Keep unknown: C stays unresolved; do not replace it with 0. → Clean record: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Preserve the original and explain each decision

AI ACADEMY / AI-1/21 · 03 · INSPECT THE LESSON MODEL

Preserve the original and explain each decision

Original record	Evidence check	Working decision
T2 height 60 cm	Card C2 says 6 cm	Correct to 6; log C2
T3 travel blank	Card C3 also blank	Keep missing; flag
T4 travel 0 cm	Card C4 confirms zero	Keep measured zero

C2 supports correcting T2. C3 cannot supply a missing measurement. C4 shows that zero is a recorded observation, not a placeholder for missing data.

Follow the labelled picture with the matching explanation. Open the vector diagram

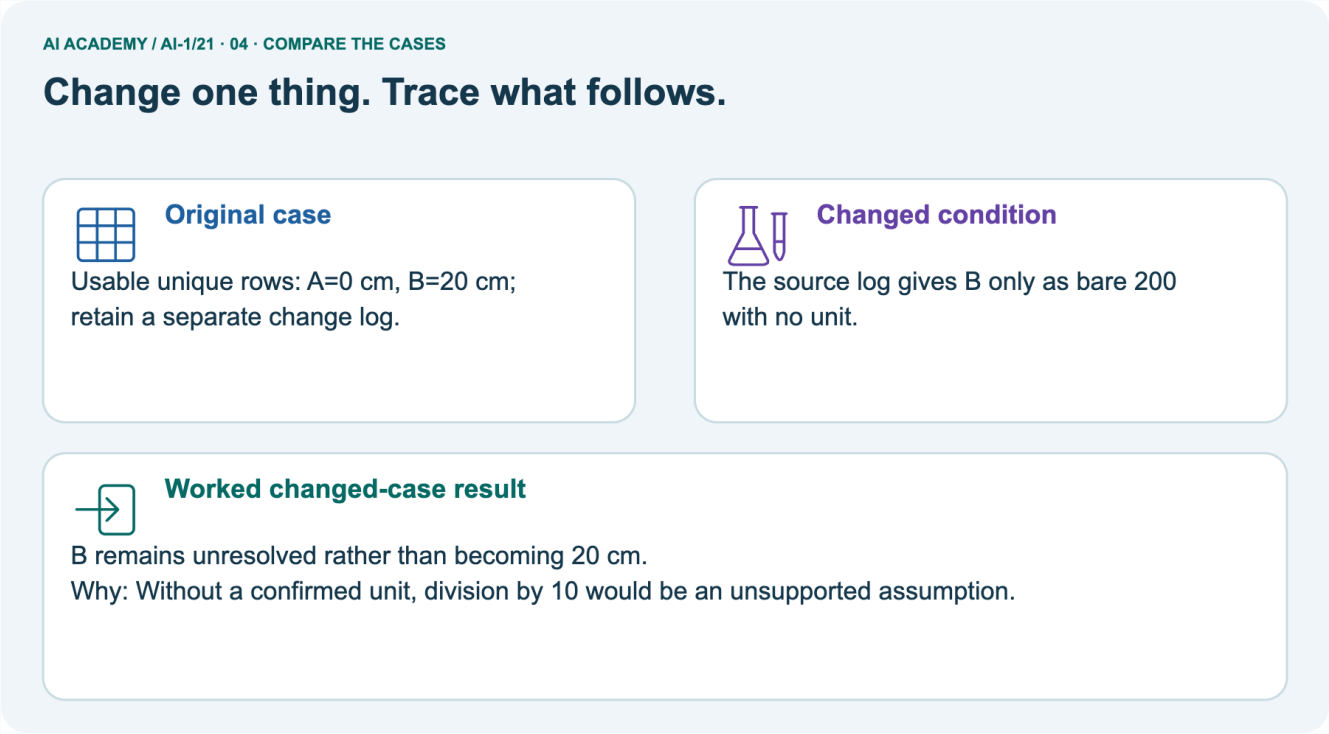
Preserve the original and explain each decision. T2 height 60 cm | Card C2 says 6 cm | Correct to 6; log C2. T3 travel blank | Card C3 also blank | Keep missing; flag.

T4 travel 0 cm | Card C4 confirms zero | Keep measured zero

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

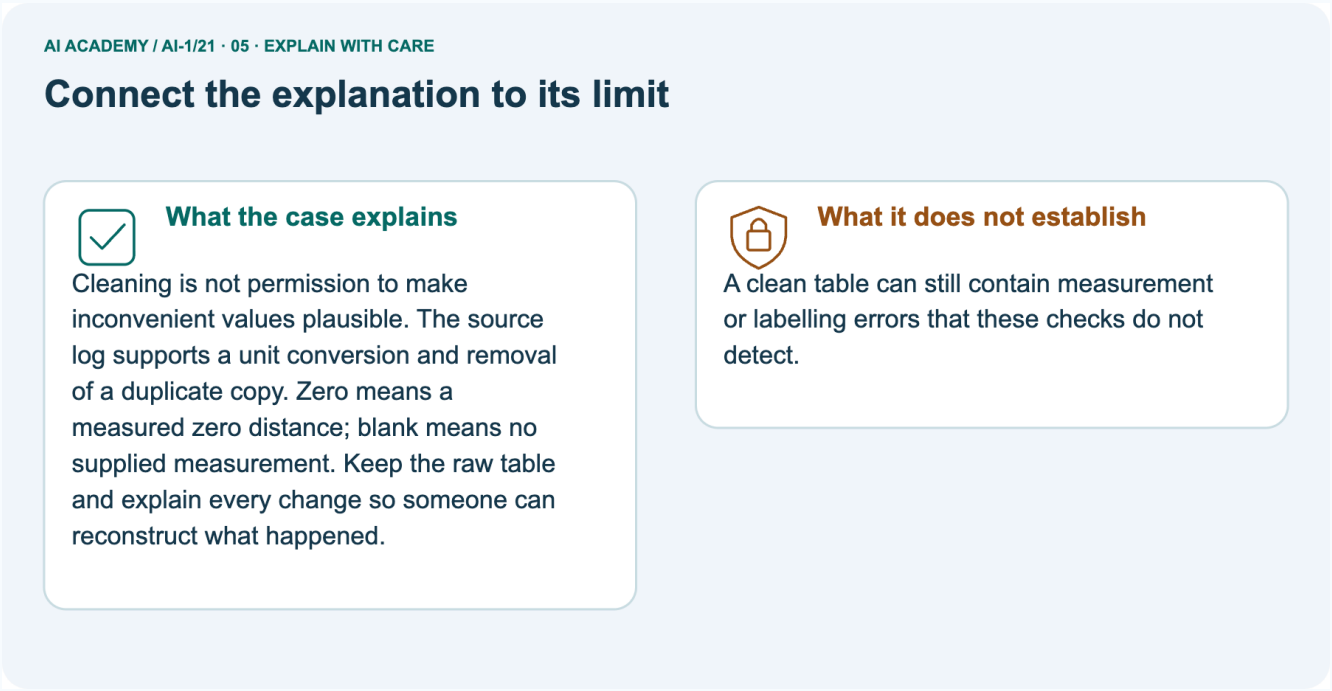
Original case: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Changed condition: The source log gives B only as bare 200 with no unit. Expected result: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

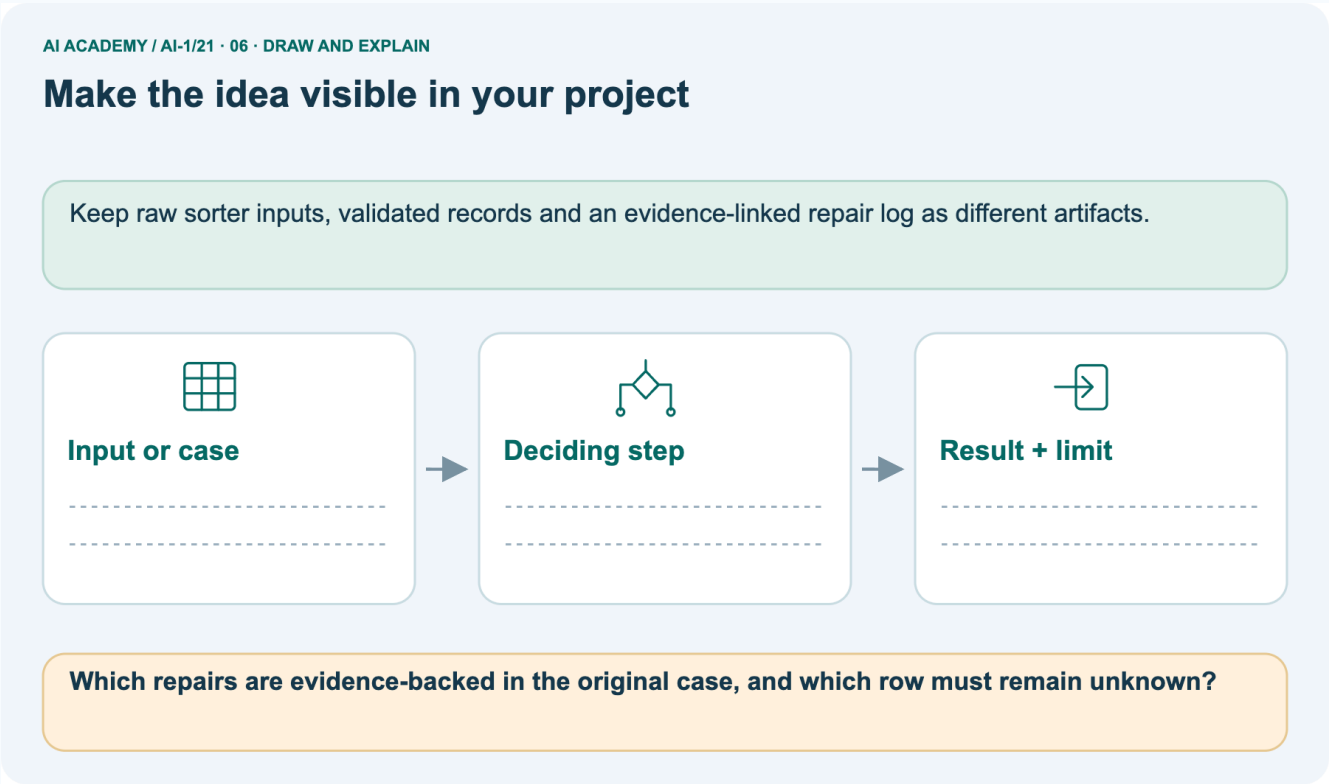
Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened. Limit: A clean table can still contain measurement or labelling errors that these checks do not detect.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts. Draw your **input**, deciding step, result and limit.

Supported question: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Good Data, Bad Data: the case

02. Follow the deciding steps

03. Preserve the original and explain each decision

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.